

Solve the parabola Equations

$$1. y^{\vee} - 8x + 8y = 0$$

$$2. y^{\vee} - 2y - 4x + 9 = 0$$

H.W.

$$3. y^{\vee} - 4y - 4x + 16 = 0$$

$$4. y^{\vee} = 4y + 4x - 8$$

$$5. y^{\vee} + 8x - 2y - 23 = 0$$

$$6. 3y^{\vee} - 10x - 12y - 18 = 0$$

$$7. x^{\vee} + 4x + 2y = 0$$

$$8. x^{\vee} - 2y - 8x + 6 = 0$$

$$9. 5x^{\vee} + 15x - 10y - 4 = 0$$

$$10. 5x^{\vee} + 30x + 2y + 59 = 0.$$

Ellipses

$$\left. \begin{array}{l} \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\ \frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \end{array} \right\} \begin{array}{l} a > b > 0 \\ 0 < e < 1 \end{array}$$

$$S = \left\{ (x, y) \mid PF + PF' = \text{constant}, \forall P \right\}$$

Eccentricity (e)

Standard form of Ellipse:

✓ Horizontal ellipse $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, a > b > 0$

Vertical ellipse $\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1, a > b > 0$

An ellipse is the set of points in a plane ^{where} the sum of whose distance from two fixed points F & F' is a constant. These two points are called foci (plural form of focus). ✓

Horizontal Ellipse:

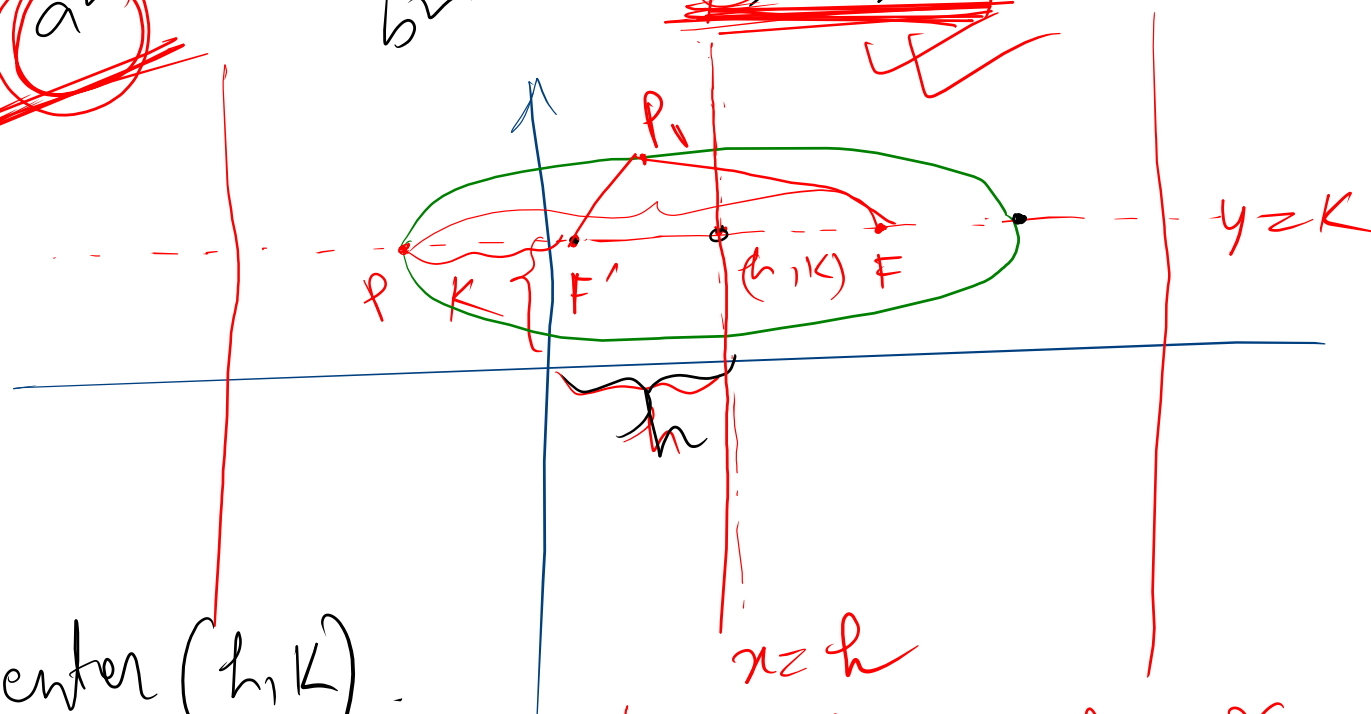
$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, \quad a > b > 0$$

$$h, k \neq 0 \quad PF + PF' = \text{constant}$$

$$P_1F + P_1F' = \text{constant}$$

$$PF + PF' = P_1F + P_1F'$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad = 2a$$



center (h, k) .

vertices $(h \pm a, k)$

Foci $(h \pm c, k)$.

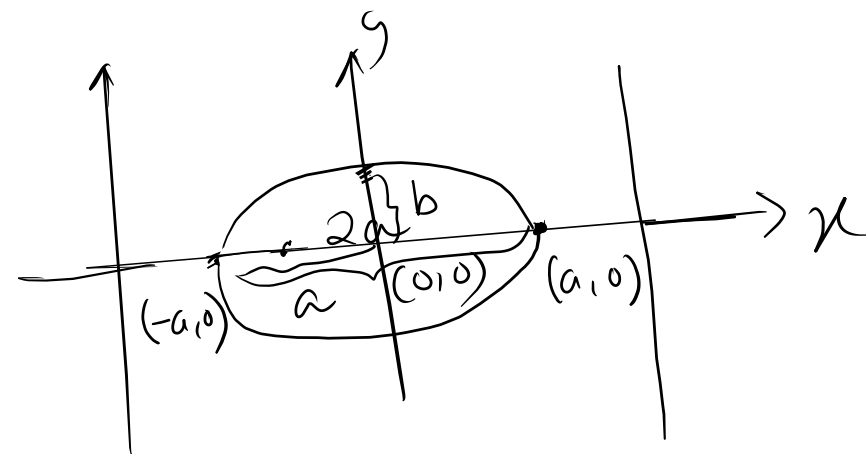
$$e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a}$$

Directrix, $x = h \pm \frac{a^2}{c}$.

length of latus rectum = $\frac{2b^2}{a}$ unit.

length of major axis $2a$

" of minor axis $2b$.

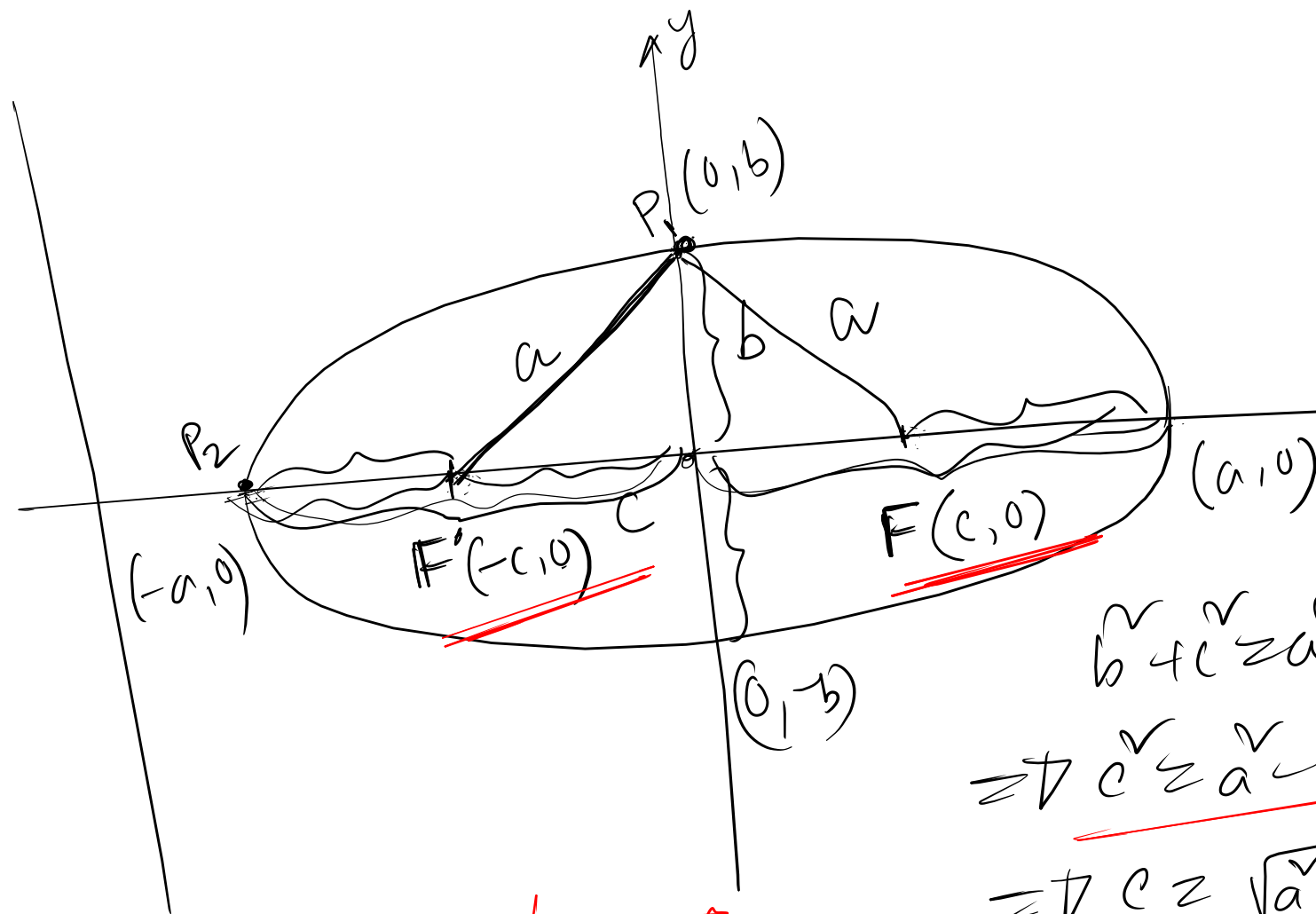


center $(0, 0)$. $c^2 = a^2 - b^2$

vertices $(\pm a, 0)$. $\Rightarrow a^2 = b^2 + c^2$

Foci $(\pm c, 0)$, $c^2 = a^2 - b^2$

Directrix, $x = \pm \frac{a^2}{c}$.



$$P_2F' + P_2F = 2a$$

$$P_1F' + P_1F = \underline{\underline{2a}}$$

$$\Rightarrow 2P_1F' = 2a$$

$$\Rightarrow P_1F' = a$$

$$b^2 + c^2 = a^2$$

$$\Rightarrow c^2 = a^2 - b^2$$

$$\Rightarrow c = \sqrt{a^2 - b^2}$$

$$0 < e = \frac{\sqrt{a^2 - b^2}}{a}, \quad [a > b]$$

$$= \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \left(\frac{b}{a}\right)^2} < 1$$

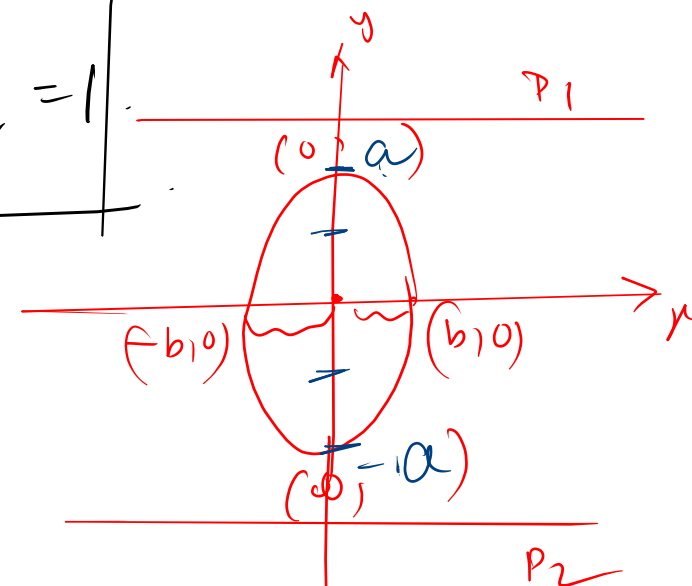
Vertical Ellipse:

$$\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$$

$$h, k > 0$$

$$a > b > 0$$

$$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1$$

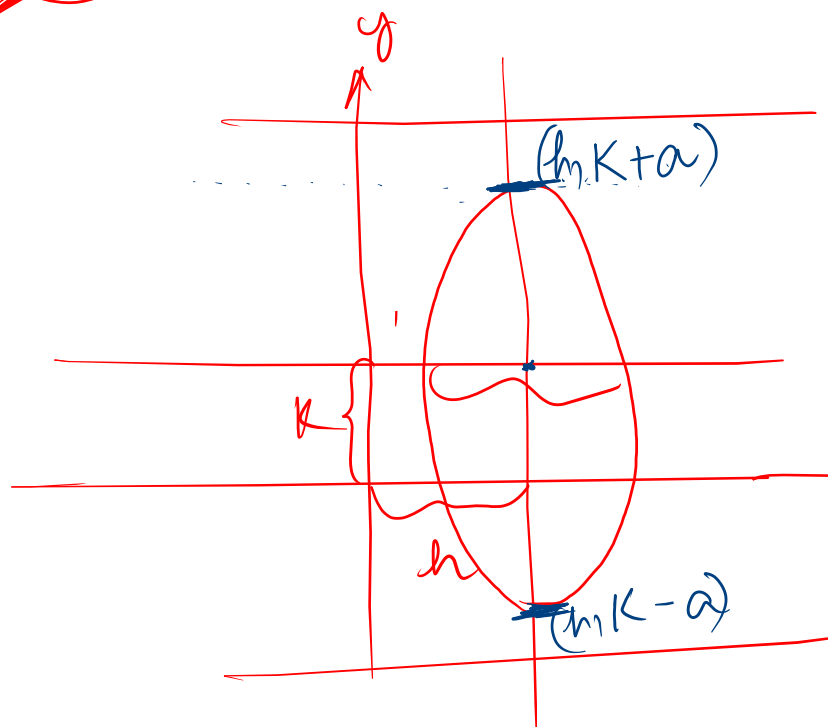


center $(0, 0)$

vertices $(0, \pm a)$

foci $(0, \pm c)$

Directrix, $y = \pm \frac{a^2}{c}$



center (h, k)

vertices $(h, k \pm a)$

foci $(h, k \pm c)$

$$c^2 = a^2 - b^2$$

$$e = \frac{c}{a}$$

Directrix, $y = k \pm \frac{a^2}{c}$

length of major axis $2a$
length of minor axis $2b$

length of major axis $2a$
length of minor axis $2b$

Transform the conic $4x^2 + 25y^2 - 50y = 75$ into standard form, find vertices, center, directrix, length of latus rectum, eccentricity, length of major axis & minor axis. & Sketch the Ellipse

Sol: $4x^2 + 25y^2 - 50y = 75$

$$\Rightarrow 4x^2 + 25(y^2 - 2y) = 75$$

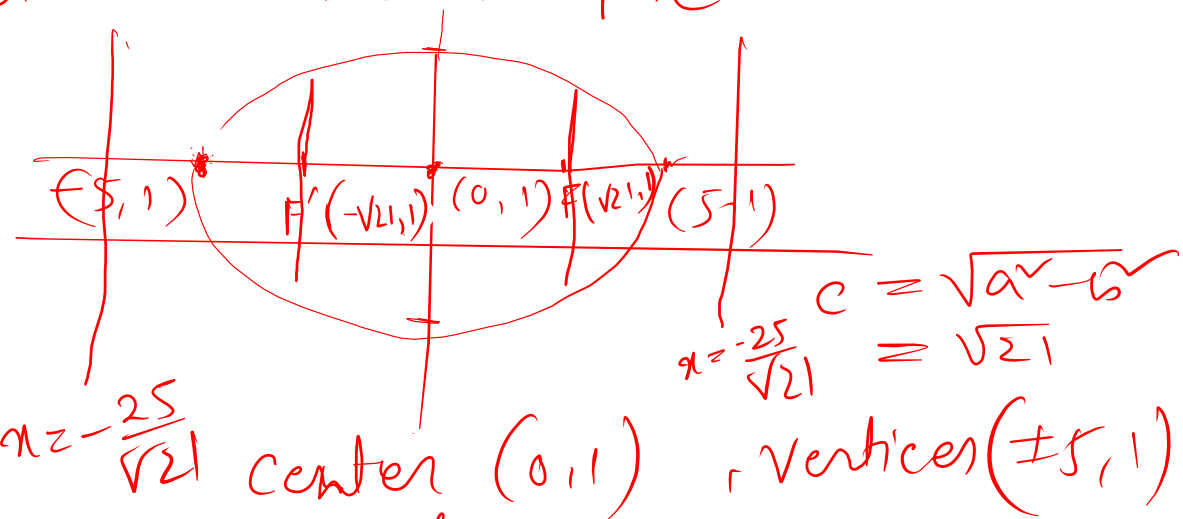
$$\Rightarrow 4x^2 + 25(y^2 - 2y + 1) = 75 + 25$$

$$\Rightarrow 4x^2 + 25(y-1)^2 = 100$$

$$\Rightarrow \frac{x^2}{25} + \frac{(y-1)^2}{4} = 1$$

$$\Rightarrow \frac{x^2}{\textcircled{5^2}} + \frac{(y-1)^2}{2^2} = 1$$

$5 > 2 > 0$
a b
a = 5, b = 2



$x = -\frac{25}{\sqrt{21}}$ center (0, 1), vertices $(\pm 5, 1)$
foci $(\pm \sqrt{21}, 1)$

Directrix, $x = \pm \frac{a^2}{c} = \pm \frac{25}{\sqrt{21}}$

$e = \frac{c}{a} = \frac{\sqrt{21}}{5}$

length of latus rectum, $\frac{2 \cdot b^2}{a} = \frac{2 \cdot 4}{5} = \frac{8}{5}$ unit